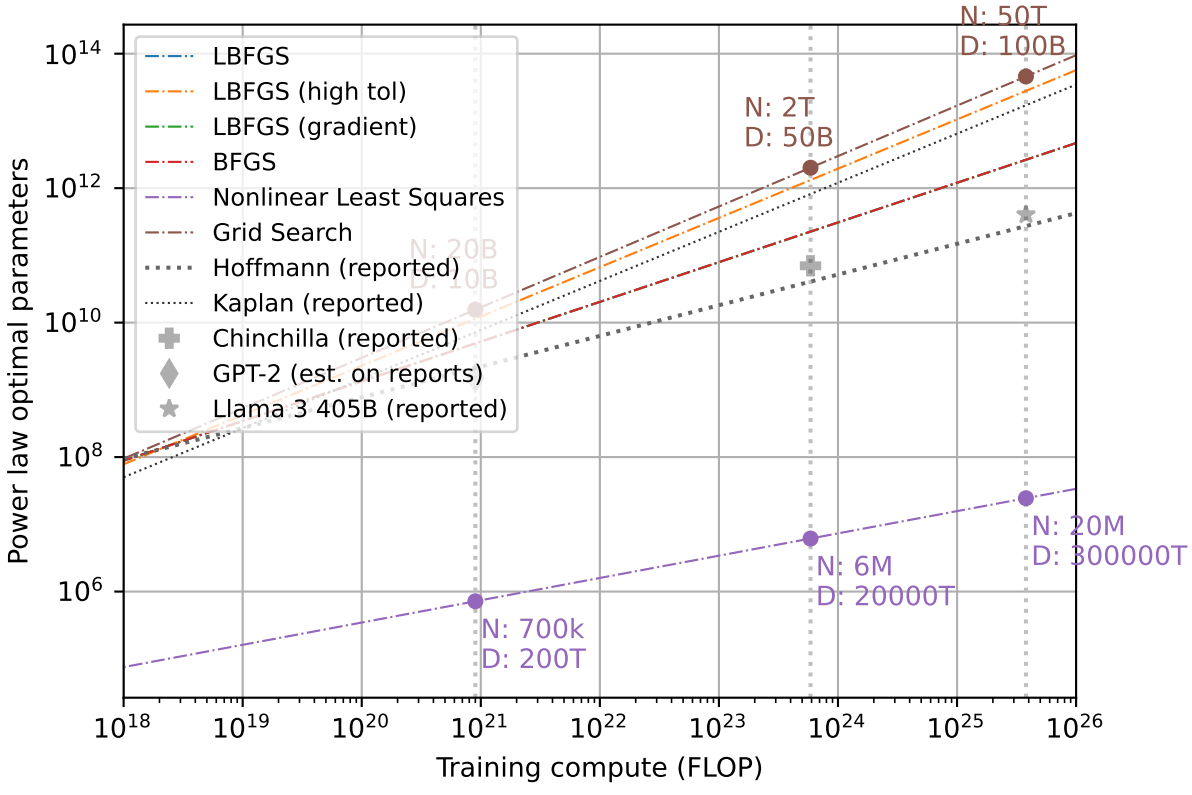
LBFGS
LBFGS (high tol)
LBFGS (gradient)
BFGS
Nonlinear Least Squares
Grid Search
Hoffmann (reported)
Kaplan (reported)
Chinchilla (reported)
GPT-2 (est. on reports)
Llama 3 405B (reported)
N: 50T
D: 100B
N: 2T
D: 50B
N: 20B
D: 10B
N: 20M
D: 300000T
N: 6M
D: 20000T
N: 700k
D: 200T
Power law optimal parameters
Training compute (FLOP)
10^{14}
10^{12}
10^{10}
10^{8}
10^{6}
10^{18}
10^{19}
10^{20}
10^{21}
10^{22}
10^{23}
10^{24}
10^{25}
10^{26}